

An Empirical Comparison of Regret-Minimizing Algorithms

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September 28, 2026

Literature: from external regret to stronger guarantees

Classical external-regret learning

Repeated-play framework: Hannan (1957)

Weighted Majority: Littlestone and Warmuth (1994)

Full information (Hedge): Freund and Schapire (1997)

Bandit feedback (EXP3): Auer et al. (2002)



Stronger regret in repeated games

Calibrated learning / CE: Foster and Vohra (1997)

Regret matching / CE: Hart and Mas-Colell (2000)

External-to-internal/swap reduction: Blum and Mansour (2007)



Stronger guarantees in structured games

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This thesis asks three questions

1. **Regret notions.** How do algorithms differ under external, internal, and swap regret at finite horizons?
2. **Robustness.** How stable are these differences across feedback models, environments, action spaces, and horizons?
3. **Repeated games.** How do regret differences appear in CE / CCE behavior, and how strongly do they depend on the opponent?

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Regret notions and equilibrium concepts

Regret notions

External regret: compare with one fixed action in hindsight.

Internal regret: replace one action i by another action j .

Swap regret: allow a separate replacement for every action i ;
that is, $i \mapsto \phi(i)$.

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CCE: no profitable fixed deviation.

CE: no profitable recommendation-dependent deviation.

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CCE: no profitable fixed deviation.

CE: no profitable recommendation-dependent deviation.

- ▶ All players external no-regret $\implies$ CCE
- ▶ All players internal / swap no-regret $\implies$ CE

Algorithms compared

Hedge

Optimistic Hedge (**OptHedge**)

Regret Matching (**RM**)

Stationary Regret Matching (**SRM**)

Blum–Mansour reduction + Hedge (**BM-Hedge**)

Blum–Mansour reduction + Optimistic Hedge
(**BM-OptHedge**)

Ito reduction + Hedge (**Ito-Hedge**)

Freund–Schapire (1997)

Chen–Peng (2020)

Hart–Mas-Colell (2000)

Hart–Mas-Colell (2000)

Blum–Mansour (2007)

Blum–Mansour (2007);
Chen–Peng (2020)

Ito (2020)

EXP3

EXP3 with implicit exploration (**EXP3-IX**)

Blum–Mansour reduction + EXP3 (**BM-EXP3**)

Ito reduction + Tsallis-INF (**Ito-Tsallis**)

Auer et al. (2002)

Neu (2015)

Blum–Mansour (2007)

Ito (2020);
Zimmert–Seldin (2021)

Huang–Pan (2023)

LCE-IX

Experimental setup

One-player environments

Historical-Frequency Adversary (**HFA**) — adaptive, non-anticipating
Lazy Random Walk (**LRW**) — oblivious, independent of the learner

Repeated games

Rock–Paper–Scissors (**RPS**)

Rock–Paper–Scissors–Lizard–Spock (**RPSLS**)

Evaluation

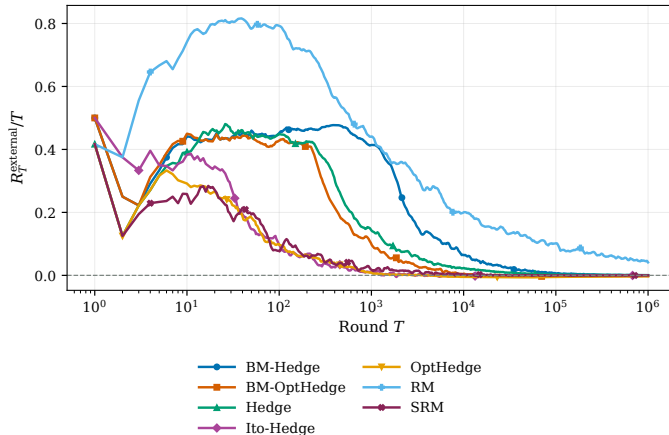
Expected external, internal, and swap action regret

CE / CCE distance in repeated games

Main trajectories: $T = 10^6$, 12 independent replicates

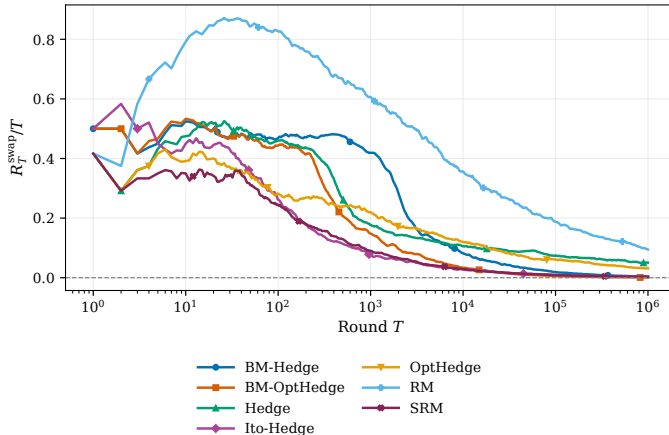
HFA: algorithm comparison across regret notions

Full information — external regret



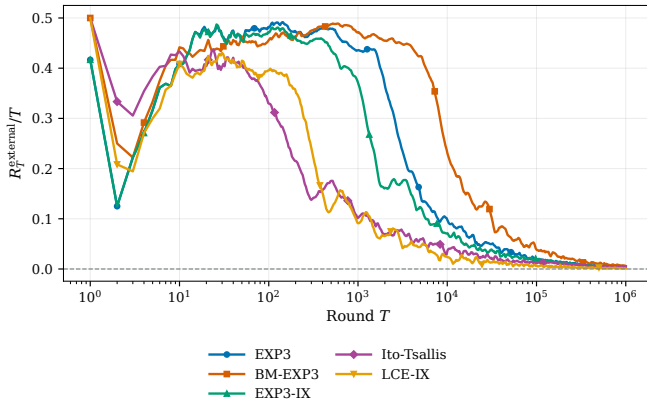
HFA: algorithm comparison across regret notions

Full information — swap regret



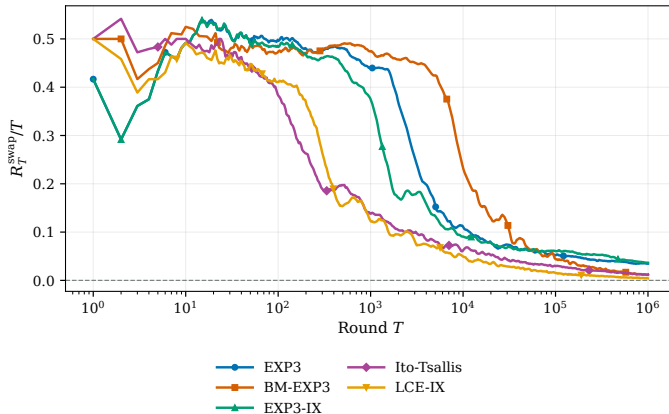
HFA: algorithm comparison across regret notions

Bandit feedback — external regret



HFA: algorithm comparison across regret notions

Bandit feedback — swap regret

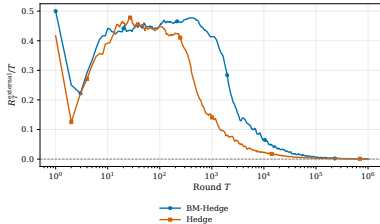


HFA: direct vs. BM-reduced learners

Full information

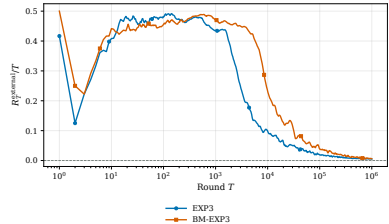
Hedge vs. BM-Hedge

Ext.

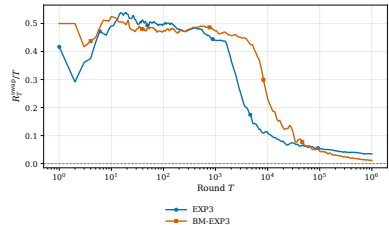
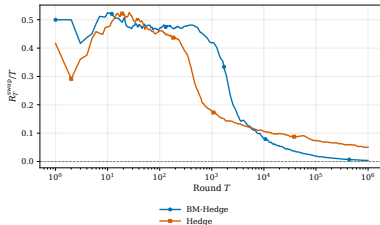


Bandit feedback

EXP3 vs. BM-EXP3



Swap

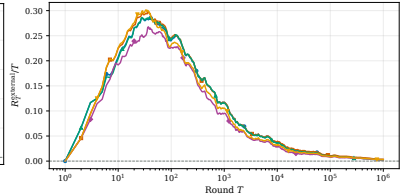
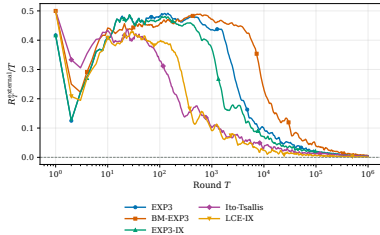


Bandit feedback: comparison across environments

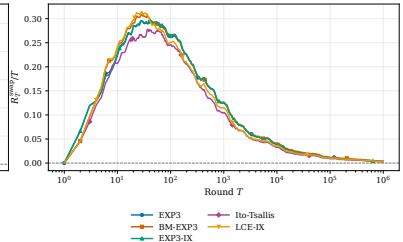
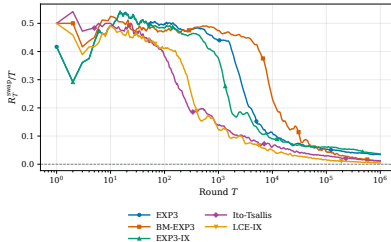
HFA

LRW

Ext.



Swap

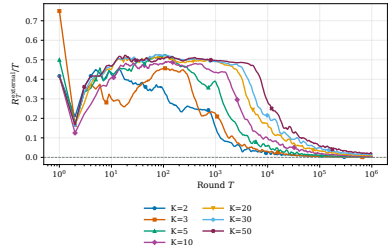
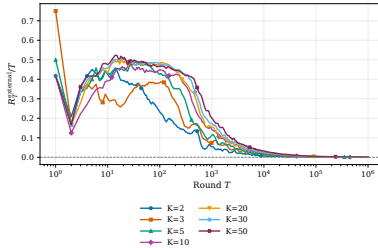


HFA: action-space scaling

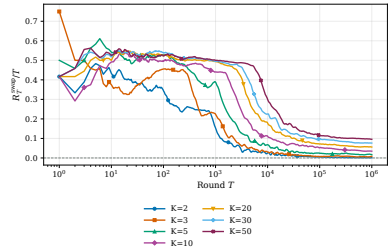
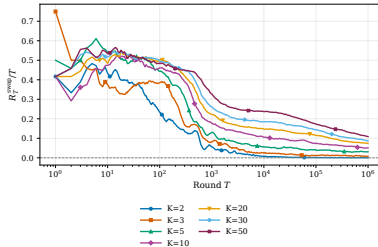
Hedge

EXP3

Ext.



Swap

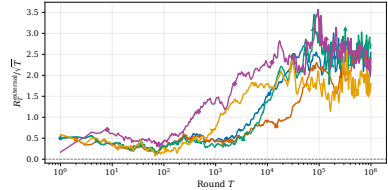
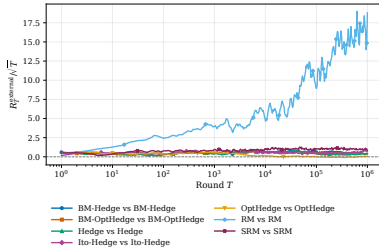


RPSLS self-play: algorithm comparison under $R_T/\sqrt{T}$

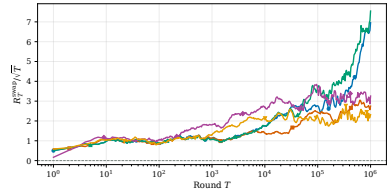
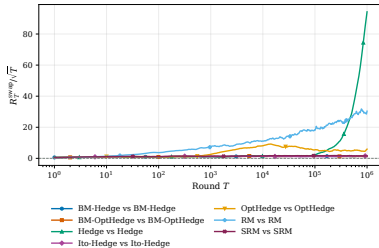
Full information

Bandit feedback

Ext.

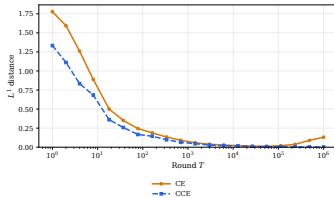
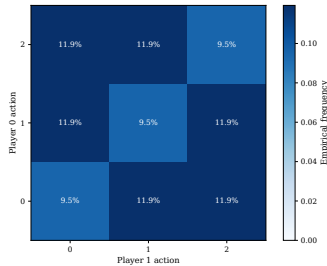


Swap

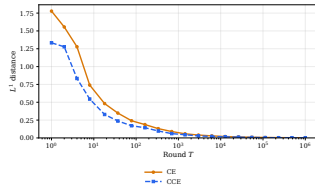
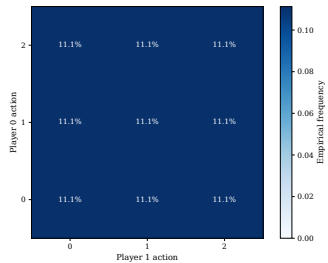


RPS self-play: joint-action structure and CE/CCE distance

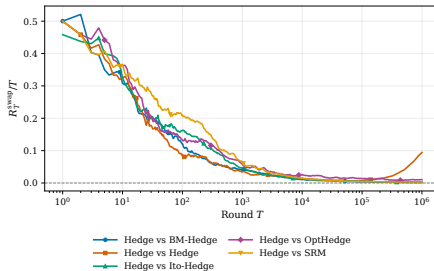
Hedge self-play



BM-Hedge self-play



RPSLS cross-play: Hedge across opponents



Hedge as player 0

Opponent	α_{ext}	α_{int}	α_{swap}
Hedge	0.474	0.915	0.964
BM-Hedge	0.479	0.516	0.510
Ito-Hedge	0.431	0.527	0.533
OptHedge	0.510	0.664	0.647
SRM	0.513	0.574	0.574

Final cumulative regret across independently configured horizons is summarized by

$$R_T \approx cT^\alpha; \alpha \text{ is a finite-horizon descriptor, not an asymptotic rate.}$$

Conclusion

1. Regret notions

Similar external-regret performance can mask substantial differences in internal and swap regret.

2. Robustness

Finite-horizon comparisons are sensitive to the reward-generation process, action-space size, and horizon.

3. Repeated games

In repeated games, CE and CCE distances can separate, and regret profiles can depend strongly on the opponent.

Thank you.

Formal action-regret definitions

For the realized action sequence, define the pairwise replacement gain

$$G_T(i, j) = \sum_{t=1}^T \mathbf{1}\{I_t = i\} (r_t(j) - r_t(i)).$$

- ▶ **External regret:** $R_T^{\text{ext}} = \max_j \sum_i G_T(i, j)$
- ▶ **Internal regret:** $R_T^{\text{int}} = \max_{i,j} G_T(i, j)$
- ▶ **Swap regret:** $R_T^{\text{swap}} = \sum_i \max_j G_T(i, j)$

$$R_T^{\text{ext}} \leq R_T^{\text{swap}}, \quad R_T^{\text{int}} \leq R_T^{\text{swap}} \leq K R_T^{\text{int}}.$$

External and internal regret are not generally ordered.

Expected action regret and nearby notions

For $q \in \{\text{ext}, \text{int}, \text{swap}\}$, the experiments report

$$\mathcal{R}_T^q = \mathbb{E}[R_T^q],$$

estimated by the mean over 12 independent replicates.

- ▶ **Action regret:** compute regret from the actions actually sampled in each run; maximize within the run before averaging.
- ▶ **Distribution regret:** evaluate the learner's mixed strategy rather than the sampled action trajectory.
- ▶ **Pseudo-regret:** the expectation and maximization are arranged differently, so it is not interchangeable with expected action regret.

Therefore, the common empirical evaluator does not directly verify every literature guarantee. *Terminology follows Greenwald et al. (2006) and Huang and Pan (2023).*

Theory guarantees and comparison caveats

Algorithm	Feedback	Regret object	Literature guarantee
Hedge	Full	external distribution	$O(\sqrt{T \log K})$
OptHedge*	Full	external distribution	$O(T^{1/6} \log^{5/6} K)$
RM / SRM**	Full	internal action	$R_T^{\text{int}} / T \rightarrow 0 \text{ a.s.}$
BM-Hedge	Full	swap distribution	$O(\sqrt{TK \log K})$
BM-OptHedge*	Full	swap distribution	$O(m^{1/2} (K \log K)^{3/4} T^{1/4})$
Ito-Hedge	Full	swap pseudo-regret	$O(\sqrt{TK \log K})$
EXP3	Bandit	external pseudo-regret	$O(\sqrt{TK \log K})$
EXP3-IX	Bandit	realized external action	$O(\sqrt{TK \log K}) \text{ w.h.p.}$
BM-EXP3	Bandit	swap pseudo-regret	$O(K \sqrt{KT \log K})$
Ito-Tsallis	Bandit	swap pseudo-regret	$O(K \sqrt{T})$
LCE-IX	Bandit	expected swap action	$O(K \sqrt{T \log K})$

* Optimistic rates require the repeated-game setting of Chen–Peng.

** RM and SRM use different convergence assumptions.

Theoretical objects and assumptions differ across rows; the empirical expected-action-regret comparison should therefore be interpreted qualitatively.

Sources: (Freund and Schapire, 1997; Chen and Peng, 2020; Hart and Mas-Colell, 2000; Blum and Mansour, 2007; Ito, 2020; Auer et al., 2002; Neu, 2015; Zimmert and Seldin, 2021; Huang and Pan, 2023).

Blum–Mansour and Ito reductions

Both construct a stochastic matrix Q_t from row learners and play a stationary distribution

$$p_t = p_t Q_t.$$

Blum–Mansour All K row learners are updated every round.

Ito Sample a row $J_t \sim p_t$ and update only the selected row learner.

The marginal played-action distribution is still p_t , because p_t is stationary.

Sources: Blum and Mansour (2007); Ito (2020).

Exact one-player reward processes

Historical-Frequency Adversary (HFA)

Before round t , actions are ranked by past play counts. The $\lceil K/2 \rceil$ most frequently played actions receive reward 0 and the rest reward 1; ties use a deterministic rotating order. HFA is adaptive but non-anticipating.

Lazy Random Walk (LRW)

Each action follows its own process on $\{0, 0.1, \dots, 1\}$, starting at 0.5. At interior states it moves down, stays, or moves up with probability $1/3$ each; at a boundary it stays or moves inward with probability $1/2$ each. Reward paths are generated independently of the learner.

Main one-player runs: $K = 10$, $T = 10^6$, 12 replicates, base seed 42. Within an LRW replicate, all compared algorithms face the same underlying reward realization.

Formal CE / CCE conditions and distance

Let μ be a distribution over joint actions.

CCE: for every player i and fixed alternative a'_i ,

$$\sum_a \mu(a) (u_i(a'_i, a_{-i}) - u_i(a)) \leq 0.$$

CE: for every player i and pair a_i, a'_i ,

$$\sum_{a_{-i}} \mu(a_i, a_{-i}) (u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i})) \leq 0.$$

$$\text{CE}(G) \subseteq \text{CCE}(G)$$

For the empirical joint-action distribution $\hat{\mu}_t$,

$$d_E(\hat{\mu}_t) = \min_{\mu \in E} \|\hat{\mu}_t - \mu\|_1, \quad E \in \{\text{CE}, \text{CCE}\}.$$

- ▶ CE / CCE constraints are linear; the experiments solve the full-dimensional L_1 projection with HiGHS.
- ▶ Distances are computed per replicate and then averaged; heatmaps average the final empirical distributions.

Finite-horizon scaling protocol

Each configured horizon is an independent experiment: the learner is reinitialized and fixed-horizon parameters are recomputed from that run's own T .

At each horizon, final cumulative action regret is averaged across replicates. Positive replicate-mean endpoints are fitted by OLS in log-log coordinates:

$$\log R_T = \alpha \log T + \log c, \quad R_T \approx cT^\alpha.$$

Interpretation: α and c are finite-horizon descriptors, not asymptotic regret rates.

Horizons:

$10^3, 2 \cdot 10^3, 3 \cdot 10^3, 5 \cdot 10^3, 10^4, 2 \cdot 10^4, 3 \cdot 10^4, 5 \cdot 10^4, 10^5, 2 \cdot 10^5, 3 \cdot 10^5, 5 \cdot 10^5, 10^6$.

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